

Problem Statement

Design a scalable video streaming platform like Netflix that allows millions of users to securely subscribe, discover, search, and stream movies and TV shows with personalized recommendations and a smooth, low-latency viewing experience.

Functional Requirements

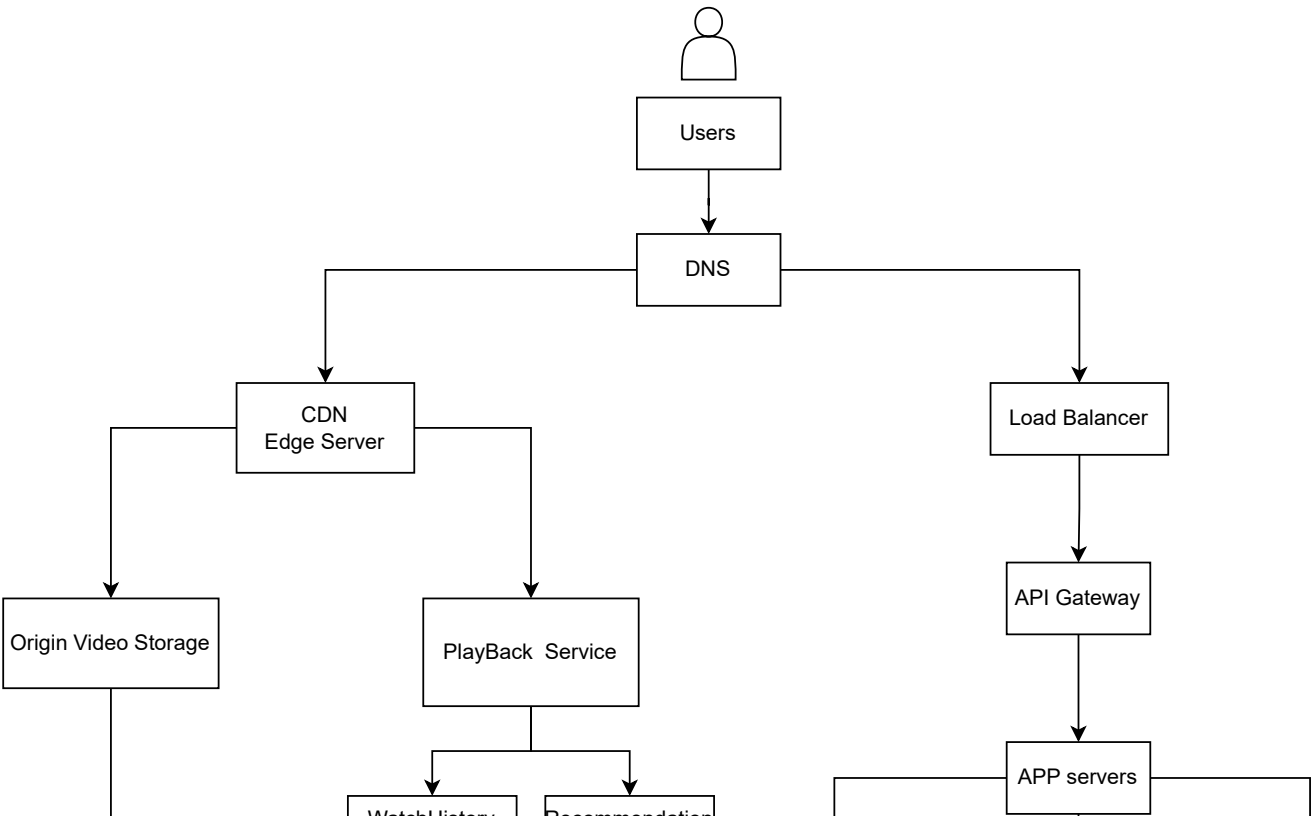
Module	Requirements
Subscription	Signup, Login, Payment, Paywall
Content	Classification, Genres, Trending, Watchlist
History	Watch History, Completion Status, Continue Watching
Search	Search Bar, Content Search, Recommendations
Player	Resolution, Playback Speed, Languages, Subtitles

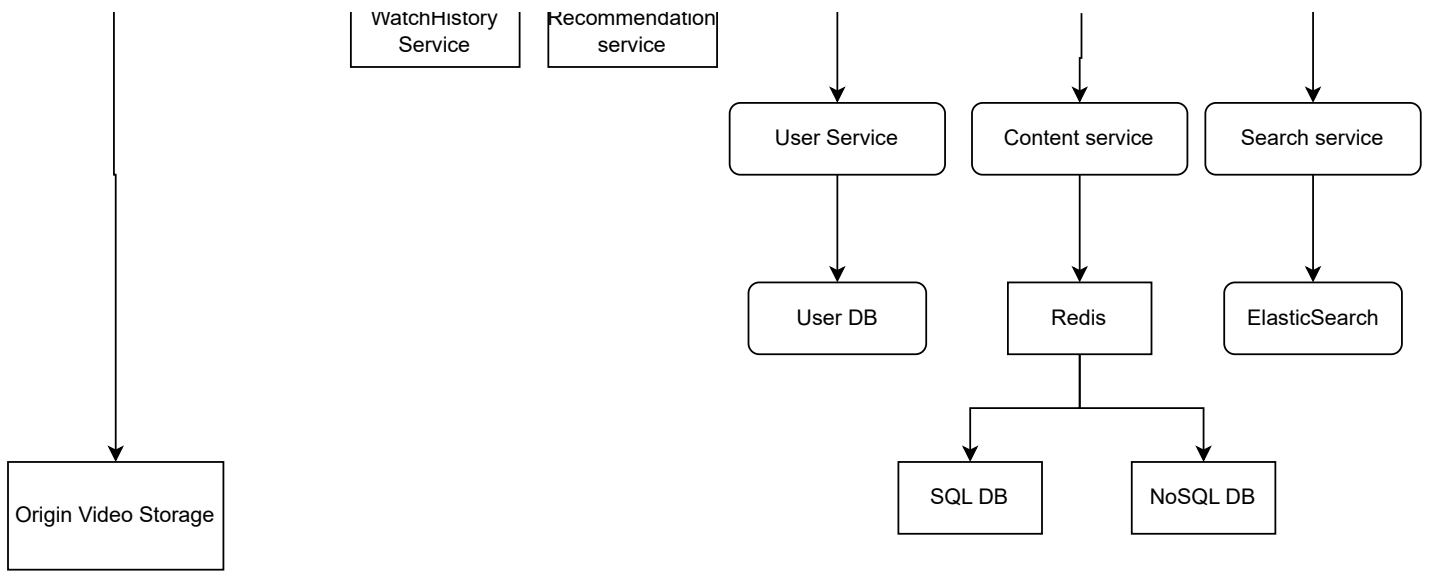
Estimations:

Metric	Estimate
Users	10M
DAU	4M
API RPS	~2.3K peak
App Servers	~10
Video Data	~8 PB/day
Peak Bandwidth	~3.75 Tbps
Origin Storage	~900 TB
CDN Cache	~60 TB
Redis	~256 GB
Watch History	~7 TB/year

Non-Functional Requirements

Requirement	Goal
Scalability	Handle millions of users
Availability	99.9%+ uptime
Low Latency	Fast API/Search response
Reliability	Smooth, low-buffer video playback
Fault Tolerance	No single point of failure
Security	Secure auth, data & payments
Durability	No loss of user/content data
Global Access	CDN-based worldwide delivery





Main Pipelines

Video Processing	Upload → Transcode → HLS/DASH → Origin Storage → CDN
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Video Playback	User → Playback Service → CDN → Video Segments → User
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Search	Content DB → Kafka → Search Indexer → Elasticsearch ↑ Search Service ↑ Search Bar
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Recommendation + History	User Activity → Kafka → T → Watch History ↳ Recommendation → Redis → Home
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